Torsional friction in planar contact

Sébastien Kleff
sk8001@nyu.edu

November 9, 2026

Abstract

Tactile sensors are widely used to detect slip, but much less is known about what can actually be inferred about frictional contact limits from tactile measurements. This question matters because detecting that slip is occurring or imminent is not the same as knowing which tangential and torsional loads the contact can still support. That distinction is important for control design. In this report, we study this question for a single planar contact patch under quasi-static Coulomb friction. We first show that coarse normal force information, consisting of the total normal force and center of pressure, is generally insufficient to determine the admissible tangential wrench set: two contacts can have the same total normal force and CoP while supporting different combinations of tangential force and torsional moment. We then study what can still be inferred from partial measurements. In particular, we derive upper and lower bounds on the admissible wrench set from coarse measurements and from spatially distributed taxel-wise normal forces. The latter provide tighter bounds by reducing the ambiguity on how the normal force is distributed over the contact patch. Taking the torques about a fixed point makes every bound a generalized moment problem: the bounds from taxel forces have closed forms, those combining taxel forces and CoP are an SOCP and an LP, and the gap between them halves, to first order, with each dyadic refinement of the taxel grid. Along a given load, the guaranteed and the possible wrench sets give a certified and an upper load factor, each a convex program, which decide whether the load is surely carried, surely not carried, or undecided by the measurement.

1 Introduction

Modern tactile sensors provide rich local information about contact pressure, shape, and deformation. Yet tactile features used for control are often selected mainly from empirical performance, for example contact pose, resultant force, image features, or local pressure measurements. A more basic question is less often made explicit: given a particular tactile measurement, what can actually be inferred about the physical interaction at the contact? We study this question in the setting of a rigid planar contact patch under Coulomb friction. Rather than trying to characterize every possible tactile signal or manipulation task, we focus on one concrete control-relevant quantity: the set of tangential forces and torsional moments that the contact can support without violating the friction model. The key difficulty is that this admissible wrench set depends on how the normal load is distributed over the patch. Coarse measurements such as the total normal force and center of pressure constrain this load distribution, but do not determine it uniquely. Spatial tactile measurements provide additional information and therefore reduce this ambiguity. The objective of this report is to make this relationship explicit. We ask **how the admissible frictional wrench set can be bounded from different levels of normal force information**, from total force and CoP to spatially distributed taxel-wise forces, **without requiring the full normal pressure field**.

2 Preliminaries (measures)

We briefly introduce the measure-theoretic notation used in the following. Only the notions needed for the contact model are recalled here; standard reference is [1].

2.1 Force as a measure.

Let $P \subset \mathbb{R}^2$ denote the contact patch, equipped with its Borel σ-algebra $\mathcal{B}(P)$. A *nonnegative measure* $\nu_N : \mathcal{B}(P) \to [0, +\infty]$ assigns to every measurable subset $A \subseteq P$ a nonnegative scalar $\nu_N(A)$, and is countably additive, i.e. for any pairwise disjoint measurable sets $A_i \subseteq P$,

$$\nu_N\left(\bigcup_i A_i\right) = \sum_i \nu_N(A_i). \tag{1}$$

In the present context, ν_N represents the (spatially distributed) normal force, i.e. $\nu_N(A)$ is the normal force carried by the region A. Hence $\nu_N(A)$ has units of force (N), and the resultant normal force is $f_N = \nu_N(P)$. We restrict attention to *finite nonnegative Radon measures*, denoted $\nu_N \in \mathcal{M}_+(P)$. Informally, a Radon measure is a "well-behaved" measure. In particular, on the compact sets considered here, it is finite and its value on measurable sets can be approximated from compact subsets. For our purpose, it essentially means that our measure does not assign an infinite force to P and it can represent either the force induced by an ordinary pressure field or an idealized concentrated point/line load.

We represent the tangential force by an $\mathbb{R}^2$-valued finite Radon measure $\nu_T = (\nu_{T,1}, \nu_{T,2})$ where each component $\nu_{T,i}$ is a finite signed Radon measure on P. Thus, for every measurable region $E \subseteq P$, the resultant tangential force carried by E is

$$\nu_T(E) = \begin{bmatrix} \nu_{T,1}(E) \\ \nu_{T,2}(E) \end{bmatrix} \in \mathbb{R}^2, \tag{2}$$

2.2 Area measure and pressure fields.

Let dA denote the usual area element on the planar patch P. A *normal pressure field* is a nonnegative function $\rho_N : P \to \mathbb{R}_+$ with units of force per unit area. If $\rho_N \in L^1_+(P)$, which simply means that ρ_N has finite integral over P

$$\rho_N(x) \geq 0 \quad \text{a.e.}, \qquad \int_P \rho_N(x)\, dA < \infty. \tag{3}$$

then it defines a finite normal force measure: the force carried by any region $E \subseteq P$ is given by

$$\nu_N(E) = \int_E \rho_N(x)\, dA. \tag{4}$$

which we also write equivalently as $d\nu_N(x) = \rho_N(x)\, dA$. In that case, ρ_N is called the *density* of the normal force measure ν_N with respect to area[1]. Not every normal force measure admits such a pressure field. For example the Dirac measure $\nu_N = f_N \delta_c$ represents an idealized normal force f_N concentrated at the single point c. It cannot be represented by an ordinary pressure field over area. In this note we adopt the measure-theoretic language (instead of only L^1_+ densities) so that we cover both "classical" pressure fields and idealized concentrated loads.

This notation allows ordinary pressure fields and concentrated loads to be handled by the same integration equations. For a measurable scalar or vector function g, the integral

$$\int_P g(x)\, d\nu_N(x) \tag{5}$$

[1] More generally, the Radon-Nikodym theorem guarantees the existence of a density for measures under suitable assumptions, such as "absolute continuity" which means physically that a subset of zero area (e.g. a point or a line) carries zero normal force. See [1], Theorem 1.30, p. 50

can be interpreted as a force-weighted spatial integral. Whenever ν_N admits a normal pressure field ρ_N,

$$\int_P g(x)\, d\nu_N(x) = \int_P g(x)\rho_N(x)\, dA. \tag{6}$$

For a Dirac measure,

$$\int_P g(x)\, d(f_N\delta_c)(x) = f_N g(c). \tag{7}$$

2.3 Moments of the normal force measure.

The zeroth moment of ν_N is its total measure,

$$m_0(\nu_N) \triangleq \int_P 1\, d\nu_N = \nu_N(P) = f_N. \tag{8}$$

In the present mechanical interpretation, this is the *resultant normal force*. The first moment is

$$m_1(\nu_N) \triangleq \int_P x\, d\nu_N(x) \in \mathbb{R}^2. \tag{9}$$

If $f_N > 0$, the *center of pressure* is the normalized first moment,

$$c[\nu_N] \triangleq \frac{m_1(\nu_N)}{m_0(\nu_N)} \tag{10}$$

$$= \frac{1}{f_N}\int_P x\, d\nu_N(x). \tag{11}$$

Hence knowing the total normal force and the center of pressure is equivalent to knowing the zeroth and first moments of the normal force measure $m_0(\nu_N) = f_N, m_1(\nu_N) = f_N c[\nu_N]$. Unlike the CoP, the pair (m_0, m_1) is linear in ν_N and defined for every measure, including $\nu_N = 0$. We work with these *homogeneous coordinates* throughout, and use the CoP only when $f_N > 0$.

Local moments. For a measurable subset $E \subseteq P$, such as a taxel, the *local first moment* of the force carried by E is $\int_E x\, d\nu_N(x)$. If $\nu_N(E) > 0$, the normalized first moment $\frac{1}{\nu_N(E)}\int_E x\, d\nu_N(x)$ is the *local CoP* of E, i.e., the barycenter of the restriction of ν_N to E; it lies in the closed convex hull of E.

3 Problem formulation

3.1 Planar contact model

Frame. All quantities are coordinates in a fixed frame $\mathcal{R} = (O, \mathcal{B})$ of the contact plane: O is the origin and $\mathcal{B} = (e_1, e_2)$ an orthonormal basis, completed by the contact normal e_3. A point of the plane is identified with its coordinate vector $x \in \mathbb{R}^2$. A wrench or a twist is always written in a frame, and its coordinates depend on the reference point (Lemma 3.4); here the reference point is O unless stated otherwise. For $x, y \in \mathbb{R}^2$, the *planar cross product* is the scalar

$$x \times y \triangleq \langle e_3, (x,0) \times (y,0)\rangle = x_1y_2 - x_2y_1 = (Jx)^\top y, \qquad J \triangleq \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}. \tag{12}$$

Let the planar contact patch $P \subset \mathbb{R}^2$ be compact, and let $\nu_N \in \mathcal{M}_+(P)$ and ν_T be the normal and tangential force measures introduced above. We do not assume $f_N > 0$. Convexity of P is assumed only where stated. We further assume a known and spatially uniform Coulomb friction

coefficient $\mu > 0$. We impose Coulomb friction in measure form by requiring that there exists a measurable density $\gamma_T : P \to \mathbb{R}^2$ of ν_T with respect to ν_N such that

$$d\nu_T(x) = \gamma_T(x)\, d\nu_N(x), \qquad \|\gamma_T(x)\|_2 \le \mu \quad \nu_N\text{-a.e. on } P. \tag{13}$$

We call γ_T the *friction ratio field*. It can be interpreted as the tangential force per unit normal force. The above condition expresses both that tangential friction force can only be carried where normal force is present and that its local magnitude is bounded by μ[2]. When ν_N admits the normal pressure field ρ_N introduced above, i.e. $d\nu_N = \rho_N\, dA$, the tangential force measure also admits a density with respect to area,

$$\rho_T(x) = \gamma_T(x)\rho_N(x), \tag{14}$$

called the *tangential traction field*. The measure-form Coulomb condition then reduces to the classical local condition

$$\|\rho_T(x)\|_2 \le \mu\rho_N(x) \qquad \text{a.e. on } P. \tag{15}$$

We define the *resultant tangential force* and the *resultant torsional moment* about the origin O as

$$f_T \triangleq \nu_T(P) = \int_P \gamma_T(x)\, d\nu_N(x), \qquad \tau_N \triangleq \int_P x \times \gamma_T(x)\, d\nu_N(x). \tag{16}$$

Definition 3.1 (Admissible tangential wrench set)**.** *For a normal force measure $\nu_N \in \mathcal{M}_+(P)$, the admissible tangential wrench set is*

$$\mathcal{X}(\nu_N) \triangleq \left\{ (f_T, \tau_N) \in \mathbb{R}^2 \times \mathbb{R} : \begin{array}{l} \exists\, \gamma_T : P \to \mathbb{R}^2 \text{ measurable such that} \\ \|\gamma_T(x)\|_2 \le \mu \quad \nu_N\text{-a.e.}, \\ f_T = \displaystyle\int_P \gamma_T(x)\, d\nu_N(x), \\ \tau_N = \displaystyle\int_P x \times \gamma_T(x)\, d\nu_N(x) \end{array} \right\}. \tag{17}$$

It is a nonempty, convex, compact subset of $\mathbb{R}^3$, symmetric about 0, containing all resultant tangential wrenches (tangential force and torsional moment about O) that are compatible with the normal force measure ν_N and the local Coulomb friction condition (13).

Proposition 3.2 (Support function of the admissible tangential wrench set)**.** *For any $\nu_N \in \mathcal{M}_+(P)$ and for every $q = (a, b) \in \mathbb{R}^2 \times \mathbb{R}$, the support function of $\mathcal{X}(\nu_N)$ is*

$$h_{\mathcal{X}(\nu_N)}(q) = \mu \int_P \|a + bJx\|_2 \; d\nu_N(x). \tag{18}$$

Proof. By definition, $\forall q = (a, b) \in \mathbb{R}^2 \times \mathbb{R}$, $h_{\mathcal{X}(\nu_N)}(q) = \sup_{(f_T,\tau_N)\in\mathcal{X}(\nu_N)} \left(a^\top f_T + b\tau_N\right)$. For any friction ratio field γ_T,

$$a^\top f_T + b\tau_N = \int_P [a + bJx]^\top \gamma_T(x)\, d\nu_N(x) \le \mu \int_P \|a + bJx\|_2 \; d\nu_N(x), \tag{19}$$

by Cauchy–Schwarz and $\|\gamma_T(x)\|_2 \le \mu$. The bound is attained by the field of magnitude μ aligned with $a + bJx$ where this vector is nonzero, and $\gamma_T^\star(x) = 0$ elsewhere. □

[2]In particular, $d\nu_T = \gamma_T\, d\nu_N$ implies $\nu_T \ll \nu_N$. Conversely, by the Radon–Nikodym theorem, a tangential measure with $\|\nu_T(E)\|_2 \le \mu\, \nu_N(E)$ for every E has such a density.

Remark 3.3 (Twists and dissipated power)**.** *The direction $q = (a, b)$ is the coordinate vector of a planar rigid twist about O: a is the velocity of the material point at O and b the angular velocity, so the point at x has velocity $v_q(x) = a + bJx$. The power of the tangential forces under this twist is $\int_P \langle v_q(x), \gamma_T(x)\rangle \, d\nu_N(x) = a^\top f_T + b\tau_N = \langle q, (f_T, \tau_N)\rangle$. Hence $h_{\mathcal{X}(\nu_N)}(q)$ is the largest power that admissible friction can exert under the twist q; since $\mathcal{X}(\nu_N) = -\mathcal{X}(\nu_N)$, it is also the maximal power that friction can dissipate when the contact slides with twist q, which is the value selected by the maximal dissipation principle. This reading requires q and the wrench to be written about the same point.*

Lemma 3.4 (Change of reference point)**.** *For $o \in \mathbb{R}^2$, let $T_o(f, \tau) \triangleq (f, \tau - o \times f)$ and $S_o(a, b) \triangleq (a + bJo, b)$ (wrench and twist coordinates about o; Varignon's formula). Then T_o and S_o are invertible linear maps and*

(i) $\langle S_o q, T_o w\rangle = \langle q, w\rangle$ for all $q, w \in \mathbb{R}^3$: the power does not depend on the reference point;

(ii) the set of wrenches about o, $\mathcal{X}_o(\nu_N) \triangleq \{(f_T, \int_P (x - o) \times \gamma_T \, d\nu_N)\}$ (over the same fields γ_T), is $T_o\mathcal{X}(\nu_N)$, and

$$h_{\mathcal{X}_o(\nu_N)}(S_o q) = h_{\mathcal{X}(\nu_N)}(q), \qquad \text{i.e.} \qquad h_{\mathcal{X}_o(\nu_N)}(a', b) = \mu \int_P \|a' + bJ(x - o)\|_2 \, d\nu_N(x). \tag{20}$$

Proof. (i) $\langle S_o q, T_o w\rangle = (a + bJo)^\top f + b(\tau - o \times f) = a^\top f + b\tau$, since $o \times f = (Jo)^\top f$. (ii) $\int_P (x - o) \times \gamma_T \, d\nu_N = \tau_N - o \times f_T$. Then $h_{\mathcal{X}_o(\nu_N)}(S_o q) = \sup_{w \in \mathcal{X}(\nu_N)} \langle S_o q, T_o w\rangle = h_{\mathcal{X}(\nu_N)}(q)$ by (i), and $S_o^{-1}(a', b) = (a' - bJo, b)$ gives the integral form. □

Remark 3.5 (Torsion about the CoP)**.** *For $f_N > 0$, the set $\mathcal{W}(\nu_N) \triangleq \mathcal{X}_{c[\nu_N]}(\nu_N)$ of wrenches about the CoP is the one usually considered for torsional friction (soft-finger models), with support function $\mu \int_P \|a + bJ(x - c[\nu_N])\|_2 \, d\nu_N$. For a single measure the two descriptions are equivalent. When several measures are combined, as in the bounds below, the change of coordinates $T_{c[\nu_N]}$ depends on the measure: unions, intersections, infima and suprema of $\mathcal{W}(\nu_N)$ over measures with different CoPs combine coordinates written about different points, which no longer describe one physical twist or one set of physical wrenches. We therefore work with $\mathcal{X}$ throughout and recover statements about the CoP through Lemma 3.4 when the CoP is measured.*

Lemma 3.6 (Linearity in the normal force)**.** *For $\nu_1, \nu_2 \in \mathcal{M}_+(P)$ and $t \geq 0$, $h_{\mathcal{X}(\nu_1 + \nu_2)} = h_{\mathcal{X}(\nu_1)} + h_{\mathcal{X}(\nu_2)}$ and $h_{\mathcal{X}(t\nu_1)} = t\, h_{\mathcal{X}(\nu_1)}$. Consequently $\mathcal{X}(\nu_1 + \nu_2) = \mathcal{X}(\nu_1) \oplus \mathcal{X}(\nu_2)$ (Minkowski sum), $\mathcal{X}(t\nu_1) = t\, \mathcal{X}(\nu_1)$ and $\mathcal{X}(0) = \{0\}$.*

Proof. The first claims follow from Proposition 3.2. The support function of a Minkowski sum is the sum of the support functions, and nonempty compact convex sets with the same support function are equal [4, Section 13]. □

3.2 Measurement-conditioned support bounds

We focus here on the case where we do not know the *true* force measure $\nu_N^\star$ but we have only access to some measurements of it (e.g. resultant normal force, CoP, etc.). The goal is to investigate what can be known about the support of $\mathcal{X}(\nu_N^\star)$ given partial measurements of $\nu_N^\star$.

Symbol	Meaning	Remark	Lives in
ν_N	normal force measure	true $\nu_N^\star$ is unknown	$\mathcal{M}_+(P)$
(m_0, m_1)	zeroth and first moments of ν_N	$m_1 = f_N c[\nu_N]$ if $f_N > 0$	$\mathbb{R} \times \mathbb{R}^2$
$q = (a, b)$	twist about O (support direction)	prescribed	$\mathbb{R}^2 \times \mathbb{R}$
$\mathcal{M}$	measurement operator	fixed by the sensing model	$\mathcal{M}_+(P) \to \mathbb{R}^r$
y	measurement value	observed data from $(\mathcal{M}, \nu_N)$	$\mathbb{R}^r$
$\Pi_{\mathcal{M}}(y)$	all measures compatible with y	depends on $(\mathcal{M}, y)$	$\subseteq \mathcal{M}_+(P)$
$\underline{h}(q; y, \mathcal{M})$	measurement-conditioned lower bound	depends on $(\mathcal{M}, y, q)$	$\mathbb{R}$
$\overline{h}(q; y, \mathcal{M})$	measurement-conditioned upper bound	depends on $(\mathcal{M}, y, q)$	$\mathbb{R}$
$\mathcal{X}^-(y, \mathcal{M})$, $\mathcal{X}^+(y, \mathcal{M})$	guaranteed and possible wrench sets	Proposition 3.11	$\subseteq \mathbb{R}^3$

Table 1: Notations.

Definition 3.7 (Measurement operator)**.** *A measurement operator is a map $\mathcal{M} : \mathcal{M}_+(P) \to \mathbb{R}^r$ of the form*

$$\mathcal{M}(\nu_N) = \begin{bmatrix} \int_P \phi_1(x)\, d\nu_N(x) \\ \vdots \\ \int_P \phi_r(x)\, d\nu_N(x) \end{bmatrix}, \qquad \forall \nu_N \in \mathcal{M}_+(P), \tag{21}$$

where the prescribed bounded Borel functions $\phi_j : P \to \mathbb{R}$ are called the measurement functions. For example, $\phi = 1$ measures the resultant normal force f_N, $\phi(x) = x_1$ and $\phi(x) = x_2$ its first moment $m_1 = f_N c$ (the CoP in homogeneous coordinates), and the indicator of a taxel the force that the taxel carries. In the following, we denote a measurement value (or observation) induced by the (unknown) true force measure $\nu_N^\star$ as $y = \mathcal{M}(\nu_N^\star) \in \mathbb{R}^r$.

Definition 3.8 (Measurement-compatible normal force family)**.** *Given a measurement operator $\mathcal{M} : \mathcal{M}_+(P) \to \mathbb{R}^r$ and the observation $y \in \mathbb{R}^r$, the set of normal force measures compatible with y is the preimage*

$$\Pi_{\mathcal{M}}(y) \triangleq \{\nu_N \in \mathcal{M}_+(P) : \mathcal{M}(\nu_N) = y\} = \mathcal{M}^{-1}[\{y\}]. \tag{22}$$

It contains all the normal force measures that are indistinguishable under the measurement y, and it is convex.

Assumption 3.9 (Mass control)**.** *There exists $\lambda \in \mathbb{R}^r$ such that $\sum_j \lambda_j \phi_j(x) \geq 1$ for every $x \in P$.*

Under Assumption 3.9, every $\nu_N \in \Pi_{\mathcal{M}}(y)$ satisfies $\nu_N(P) \leq \int_P \sum_j \lambda_j \phi_j \, d\nu_N = \lambda^\top y$, so the compatible family has bounded mass. All measurement operators studied below measure the resultant normal force, directly or as the sum of the taxel forces, and satisfy it. Without it, force could be hidden where all the measurement functions vanish, and the upper bound below would be infinite. No positivity assumption on f_N is needed.

Given an observation y of the unknown true force measure $\nu_N^\star$, we can compute measurement-conditioned lower/upper bounds on the support function $h_{\mathcal{X}(\nu_N^\star)}$ by optimizing over $\Pi_{\mathcal{M}}(y)$.

Definition 3.10 (Measurement-conditioned support bounds)**.** *Let $\mathcal{M}$ be a measurement operator and $y \in \mathbb{R}^r$ an observation. For a twist $q \in \mathbb{R}^2 \times \mathbb{R}$, the measurement-conditioned bounds in direction q are*

$$\underline{h}(q; y, \mathcal{M}) \triangleq \inf_{\nu_N \in \Pi_{\mathcal{M}}(y)} h_{\mathcal{X}(\nu_N)}(q) = \inf_{\nu_N \in \Pi_{\mathcal{M}}(y)} \mu \int_P \|a + bJx\|_2 \, d\nu_N(x), \tag{23}$$

$$\overline{h}(q; y, \mathcal{M}) \triangleq \sup_{\nu_N \in \Pi_{\mathcal{M}}(y)} h_{\mathcal{X}(\nu_N)}(q) = \sup_{\nu_N \in \Pi_{\mathcal{M}}(y)} \mu \int_P \|a + bJx\|_2 \, d\nu_N(x), \tag{24}$$

and the guaranteed and possible wrench sets are

$$\mathcal{X}^-(y,\mathcal{M}) \triangleq \bigcap_{\nu_N \in \Pi_{\mathcal{M}}(y)} \mathcal{X}(\nu_N), \qquad \mathcal{X}^+(y,\mathcal{M}) \triangleq \bigcup_{\nu_N \in \Pi_{\mathcal{M}}(y)} \mathcal{X}(\nu_N). \tag{25}$$

The objective of both problems in (23) is linear in ν_N, and the constraints $\int_P \phi_j \, d\nu_N = y_j$ are linear moment constraints: *the bounds are Generalized Moment Problems (GMP)* [2], for every measurement operator. This is a consequence of the fixed reference point; about the CoP of each measure the objective would depend on ν_N through $c[\nu_N]$ (Remark 3.5).

Proposition 3.11 (Guaranteed and possible wrench sets)**.** *Let $\mathcal{M}$ satisfy Assumption 3.9 and $y \in \mathbb{R}^r$.*

(i) The possible set $\mathcal{X}^+(y,\mathcal{M})$ is convex, and its support function is $\overline{h}(\cdot; y, \mathcal{M})$.

(ii) The guaranteed set is $\mathcal{X}^-(y,\mathcal{M}) = \{w : \langle q, w\rangle \le \underline{h}(q; y, \mathcal{M}) \ \forall q\}$. Its support function is the closed convex envelope of $\underline{h}(\cdot; y, \mathcal{M})$, which is in general strictly below $\underline{h}$.

(iii) $y \mapsto \overline{h}(q; y, \mathcal{M})$ is concave and $y \mapsto \underline{h}(q; y, \mathcal{M})$ is convex on the set of observations $\mathcal{M}(\mathcal{M}_+(P))$, and both are positively homogeneous.

Proof. (i) Let $w_k \in \mathcal{X}(\nu_k)$ with $\nu_k \in \Pi_{\mathcal{M}}(y)$, $k = 1, 2$, and $t \in [0, 1]$. By Lemma 3.6, $tw_1 + (1-t)w_2 \in t\mathcal{X}(\nu_1) \oplus (1-t)\mathcal{X}(\nu_2) = \mathcal{X}(t\nu_1 + (1-t)\nu_2)$, and $t\nu_1 + (1-t)\nu_2 \in \Pi_{\mathcal{M}}(y)$ since $\Pi_{\mathcal{M}}(y)$ is convex. The support function of a union is the supremum of the support functions. (ii) $w \in \mathcal{X}^-$ if and only if $w \in \mathcal{X}(\nu_N)$ for every compatible ν_N, i.e., $\langle q, w\rangle \le h_{\mathcal{X}(\nu_N)}(q)$ for all q and all compatible ν_N [4, Section 13]. The support function of the set $\{w : \langle q, w\rangle \le \underline{h}(q) \ \forall q\}$ is the closed convex envelope of $\underline{h}$ [4, Section 13]. (iii) If $\nu_k \in \Pi_{\mathcal{M}}(y_k)$, then $t\nu_1 + (1-t)\nu_2 \in \Pi_{\mathcal{M}}(ty_1 + (1-t)y_2)$, and $\nu_N \mapsto h_{\mathcal{X}(\nu_N)}(q)$ is linear; take the supremum (resp. infimum) over ν_1, ν_2. □

Property (i) depends on the fixed reference point: with the CoP-referenced sets $\mathcal{W}(\nu_N)$, a convex combination of measures with different CoPs moves the reference point, and the union need not be convex. Property (iii) generalizes the concavity of Proposition 4.4 of Yann's draft (Remark 4.6).

Remark 3.12 (How to use the bounds)**.** *A candidate tangential wrench w (about O) is carried by every compatible normal force measure, i.e., $w \in \mathcal{X}^-(y,\mathcal{M})$, if and only if $\langle q, w\rangle \le \underline{h}(q; y, \mathcal{M})$ for all q. Conversely, if $\langle q, w\rangle > \overline{h}(q; y, \mathcal{M})$ for some q, then $w \notin \mathcal{X}^+(y,\mathcal{M})$: no compatible measure carries w, and the contact slips whatever the actual pressure distribution. By Proposition 3.11(i), the second test is a convex membership test. The sets are closed: a wrench with $\langle q, w\rangle = h_{\mathcal{X}(\nu_N)}(q)$ for some q lies on the boundary of $\mathcal{X}(\nu_N)$, where the contact is at the onset of sliding with twist q (maximal dissipation). A strict inequality, or a margin, certifies the interior. Section 9 turns both tests into convex programs along a load direction*[3]*.*

3.3 General structure of the bounds

For $q = (a, b) \in \mathbb{R}^2 \times \mathbb{R}$, we write

$$g(x; q) \triangleq \|a + bJx\|_2 \,, \tag{26}$$

the speed of the material point at x under the twist q, so that, by Proposition 3.2, $h_{\mathcal{X}(\nu_N)}(q) = \mu \int_P g(x; q) \, d\nu_N(x)$. For $b \neq 0$, $a + bJx = bJ(x - p)$ with $p \triangleq Ja/b$, the instantaneous center of rotation, so that $g(x; q) = |b| \, \|x - p\|_2$. For $b = 0$, $g(x; q) = \|a\|_2$.

[3]Because all support functions are positively homogeneous in q, it is sufficient to consider one normalized representative of each nonzero ray.

Lemma 3.13 (Properties of the integrand). *For every $q = (a, b) \in \mathbb{R}^2 \times \mathbb{R}$:*

(i) *$x \mapsto g(x; q)$ is convex on $\mathbb{R}^2$;*

(ii) *for all $x, x' \in \mathbb{R}^2$, $|g(x; q) - g(x'; q)| \le |b| \, \|x - x'\|_2$;*

(iii) *for every $\nu_N \in \mathcal{M}_+(P)$,*

$$h_{\mathcal{X}(\nu_N)}(q) \ge \mu \, \|m_0(\nu_N)\, a + bJm_1(\nu_N)\|_2 \,, \tag{27}$$

which, for $f_N > 0$, reads $h_{\mathcal{X}(\nu_N)}(q) \ge \mu f_N \, g(c[\nu_N]; q)$: the maximal dissipated power is at least that of the whole normal force concentrated at the CoP.

Proof. (i) $g(\cdot; q)$ is a norm composed with an affine map. (ii) By the reverse triangle inequality and $\|J\|_2 = 1$. (iii) By the triangle inequality for vector-valued integrals, $\int_P \|a + bJx\|_2 \, d\nu_N \ge \| \int_P (a + bJx) \, d\nu_N \|_2 = \|m_0 a + bJm_1\|_2$. □

Comparing measurement operators. The bounds of two measurement operators can only be compared if their observations come from the same normal force measure. This leads to an order on the information carried by measurement operators.

Definition 3.14 (Information preorder). *A measurement operator $\mathcal{M}$ is at least as informative as $\mathcal{M}'$, written $\mathcal{M} \succeq \mathcal{M}'$, if*

$$\Pi_{\mathcal{M}}(\mathcal{M}(\nu_N)) \subseteq \Pi_{\mathcal{M}'}(\mathcal{M}'(\nu_N)) \qquad \text{for every } \nu_N \in \mathcal{M}_+(P), \tag{28}$$

i.e., every measure that $\mathcal{M}$ cannot distinguish from ν_N is also indistinguishable from ν_N under $\mathcal{M}'$. The concatenation $\mathcal{M} \vee \mathcal{M}' \triangleq (\mathcal{M}, \mathcal{M}')$ is the operator with measurement functions $(\phi_1, \dots, \phi_r, \phi'_1, \dots, \phi'_{r'})$.

Proposition 3.15 (Properties of the information preorder). *(i) $\succeq$ is a preorder (reflexive and transitive). It is not antisymmetric, and not total.*

(ii) *$\mathcal{M} \succeq \mathcal{M}'$ if and only if there is a matrix $A \in \mathbb{R}^{r' \times r}$ with $\phi'_j(x) = \sum_k A_{jk} \phi_k(x)$ for every $x \in P$ and every j, i.e., $\mathcal{M}' = A\,\mathcal{M}$.*

(iii) *$\mathcal{M} \vee \mathcal{M}' \succeq \mathcal{M}$ and $\mathcal{M} \vee \mathcal{M}' \succeq \mathcal{M}'$, with $\Pi_{\mathcal{M} \vee \mathcal{M}'}((y, y')) = \Pi_{\mathcal{M}}(y) \cap \Pi_{\mathcal{M}'}(y')$.*

Proof. (i) Inclusion is reflexive and transitive. Scaling the measurement functions by 2 gives a different operator with the same compatible families, and the operators $\mathcal{M}_0$ and $\mathcal{M}_F$ of the next sections are not comparable (see (ii)). (ii) If $\mathcal{M}' = A\mathcal{M}$, then $\mathcal{M}(\nu) = \mathcal{M}(\nu_N)$ implies $\mathcal{M}'(\nu) = \mathcal{M}'(\nu_N)$. Conversely, write $\Phi(x) = (\phi_k(x))_k$ and $\Phi'(x) = (\phi'_j(x))_j$. A linear map A with $A\Phi(x) = \Phi'(x)$ for all $x \in P$ exists if and only if $\sum_i \alpha_i \Phi(x_i) = 0$ implies $\sum_i \alpha_i \Phi'(x_i) = 0$ for all finite families $x_i \in P$, $\alpha_i \in \mathbb{R}$. Given such a family, let $\sigma^{\pm} = \sum_{i : \pm\alpha_i > 0} |\alpha_i| \delta_{x_i} \in \mathcal{M}_+(P)$. Then $\mathcal{M}(\sigma^+) = \mathcal{M}(\sigma^-)$, so $\sigma^- \in \Pi_{\mathcal{M}}(\mathcal{M}(\sigma^+)) \subseteq \Pi_{\mathcal{M}'}(\mathcal{M}'(\sigma^+))$, i.e., $\sum_i \alpha_i \Phi'(x_i) = \mathcal{M}'(\sigma^+) - \mathcal{M}'(\sigma^-) = 0$. (iii) Immediate from the definitions. □

The characterization (ii) is pointwise: functions that agree almost everywhere with respect to area are not equivalent, since point loads see their values at single points. The number r of measurement functions is not monotone with respect to $\succeq$, because measurement functions can be redundant. The operator without measurement functions ($r = 0$, $\Pi = \mathcal{M}_+(P)$) is below every operator; its bounds are 0 and $+\infty$, and it violates Assumption 3.9.

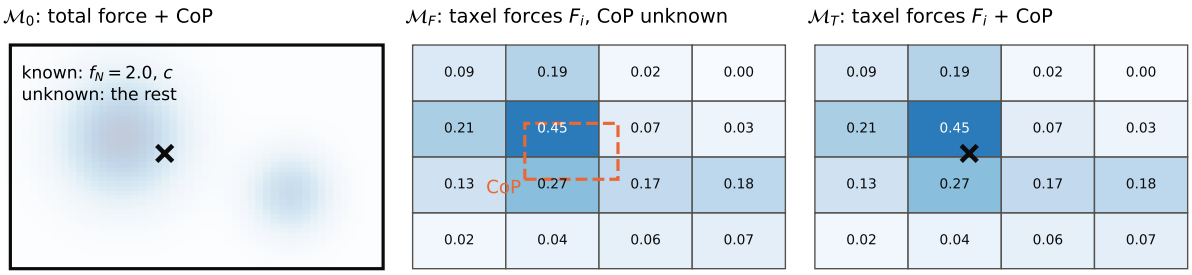


Figure 1: What each measurement operator observes for the same contact (4×4 taxels). $\mathcal{M}_0$: only f_N and the CoP, the pressure distribution (faded) is unknown. $\mathcal{M}_F$: the taxel forces F_i (cells filled with the mean pressure $F_i/|P_i|$), the CoP is only known to lie in a taxel-shaped rectangle centered at the force-weighted average of the taxel centers (dashed, Remark 5.8). $\mathcal{M}_T$: taxel forces and CoP.

Measurement	Observed data	Lower bound	Upper bound
$\mathcal{M}_0$ (Sec. 4)	$y_0 = (f_N, m_1)$	closed form	LP over the vertices of P (closed form via triangles)
$\mathcal{M}_F$ (Sec. 6)	$y_F = (F_1, \dots, F_r)$	closed form: $\sum_i$ of minima over the taxels	closed form: $\sum_i$ of maxima over the taxel corners
$\mathcal{M}_T$ (Sec. 7)	$y_T = (y_0, y_F)$	SOCP	LP over the taxel corners

Table 2: Measurement operators and structure of the resulting bounds. All bounds are GMPs; the CoP constraint of $\mathcal{M}_T$ couples the taxels.

Theorem 3.16 (Monotonicity with respect to information)**.** *Let $\mathcal{M} \succeq \mathcal{M}'$, $\nu_N \in \mathcal{M}_+(P)$, $y = \mathcal{M}(\nu_N)$ and $y' = \mathcal{M}'(\nu_N)$. Then, for every $q \in \mathbb{R}^2 \times \mathbb{R}$,*

$$\underline{h}(q; y', \mathcal{M}') \le \underline{h}(q; y, \mathcal{M}) \le h_{\mathcal{X}(\nu_N)}(q) \le \overline{h}(q; y, \mathcal{M}) \le \overline{h}(q; y', \mathcal{M}'), \tag{29}$$

and $\mathcal{X}^-(y', \mathcal{M}') \subseteq \mathcal{X}^-(y, \mathcal{M}) \subseteq \mathcal{X}(\nu_N) \subseteq \mathcal{X}^+(y, \mathcal{M}) \subseteq \mathcal{X}^+(y', \mathcal{M}')$. This applies in particular to the true measure $\nu_N^\star$.

Proof. $\nu_N \in \Pi_{\mathcal{M}}(y) \subseteq \Pi_{\mathcal{M}'}(y')$. An infimum (resp. supremum, intersection, union) over a smaller family is larger (resp. smaller, larger, smaller). □

Remark 3.17 (Attainment)**.** *The bounds are defined as an infimum and a supremum, and we do not need them to be attained: only their values enter the tests of Remark 3.12 and the monotonicity of Theorem 3.16. For the taxel measurements, the finite-dimensional formulas below are attained by measures that may put force on a taxel boundary owned by a neighboring taxel; Lemma 5.5 shows that they are nevertheless the exact values of the bounds.*

Table 2 summarizes the measurement operators studied in the remainder of the report and the structure of the resulting problems, and Figure 1 illustrates what each of them observes. We next specialize (23) by increasing information: the coarse measurement (Section 4), the taxel forces (Section 6), and both (Section 7). The tightening of the bounds as information is added is studied in Section 8.

4 $\mathcal{M}_0$ case (coarse measurement)

We first assume that the available measurements consist only of the resultant normal force and its first moment, i.e., the CoP when $f_N > 0$. **Known:** P, μ, f_N and m_1. **Unknown:** how $\nu_N^\star$ is distributed over P. In this section P is convex.

Definition 4.1 (Coarse normal force measurement)**.** *Given the true (unknown) normal force measure $\nu_N^\star \in \mathcal{M}_+(P)$, the coarse measurement is its zeroth and first moments*

$$y_0 = \mathcal{M}_0(\nu_N^\star) \triangleq \left(\int_P d\nu_N^\star(x),\ \int_P x\, d\nu_N^\star(x) \right) = (f_N, m_1). \tag{30}$$

Its measurement functions are $\phi_1(x) = 1$, $\phi_2(x) = x_1$, $\phi_3(x) = x_2$, and it satisfies Assumption 3.9 with $\lambda = (1, 0, 0)$. For $f_N > 0$, y_0 is equivalent to (f_N, c) with $c = m_1/f_N \in P$.

Theorem 4.2 (Exact coarse lower bound)**.** *Given the coarse observation $y_0 = (f_N, m_1)$, for any $q = (a, b) \in \mathbb{R}^2 \times \mathbb{R}$,*

$$\boxed{\underline{h}(q; y_0, \mathcal{M}_0) = \mu \, \| f_N a + b J m_1 \|_2} \tag{31}$$

For $f_N > 0$ this is $\mu f_N \, g(c; q)$, the maximal dissipated power of the normal force concentrated at the CoP.

Proof. By Lemma 3.13(iii), $h_{\mathcal{X}(\nu_N)}(q) \geq \mu \| f_N a + b J m_1 \|_2$ for every $\nu_N \in \Pi_{\mathcal{M}_0}(y_0)$. If $f_N = 0$, then $\Pi_{\mathcal{M}_0}(y_0) = \{0\}$ and both sides vanish. Otherwise $c \in P$, so $f_N \delta_c \in \Pi_{\mathcal{M}_0}(y_0)$, and it attains the value. □

Corollary 4.3 (Exact guaranteed set; no torsion about the CoP)**.** *Let $f_N > 0$ and $c = m_1/f_N$. Then*

$$\mathcal{X}^-(y_0, \mathcal{M}_0) = \mathcal{X}(f_N \delta_c) = \{(f, c \times f) : \ \|f\|_2 \leq \mu f_N\}. \tag{32}$$

In coordinates about the CoP (Lemma 3.4), $T_c \mathcal{X}^-(y_0, \mathcal{M}_0) = f_N \, B(0, \mu) \times \{0\}$: the coarse measurement guarantees the friction disc for the tangential force, but no torsional moment about the CoP. Equivalently, for the twists rotating about the CoP, $q = (-bJc, b)$, $\underline{h}(q; y_0, \mathcal{M}_0) = 0$.

Proof. For the point load, a field is a single vector $\gamma \in B(0, \mu)$, with resultant $(f_N \gamma,\ c \times f_N \gamma)$, which gives the second equality, and $\mathcal{X}^- \subseteq \mathcal{X}(f_N \delta_c)$ since $f_N \delta_c$ is compatible. Conversely, let $\|f\|_2 \leq \mu f_N$ and $\nu_N \in \Pi_{\mathcal{M}_0}(y_0)$. The uniform field $\gamma_T \equiv f/f_N$ is admissible, with $f_T = f$ and $\tau_N = \int_P x \times (f/f_N)\, d\nu_N = m_1 \times f/f_N = c \times f$ (compare Remark 4.2 of Yann's draft). □

Theorem 4.4 (Coarse upper bound)**.** *The supremum $\overline{h}(q; y_0, \mathcal{M}_0)$ is attained by a measure $\nu_N^+ \in \Pi_{\mathcal{M}_0}(y_0)$ supported on at most three points, i.e.,*

$$\nu_N^+ = \sum_{j=1}^{s} \lambda_j \delta_{x_j}, \qquad s \leq 3, \qquad x_j \in P, \quad \lambda_j \geq 0, \qquad \sum_{j=1}^{s} \lambda_j = f_N, \qquad \sum_{j=1}^{s} \lambda_j x_j = m_1. \tag{33}$$

Proof. This is Proposition 4.4 of Yann's draft with the Coulomb set $C = B(0, \mu)$ (Remark 4.6), where it follows from the fact that the extreme points of a set of measures defined by three linear moment constraints are supported on at most three points [3, 2]. A finite-dimensional argument is as follows. If $f_N = 0$ there is nothing to prove. Otherwise, the constraints and the objective depend on $\pi = \nu_N / f_N$ only through its moment vector $\int_P \Phi \, d\pi$, with $\Phi(x) = (x, g(x; q)) \in \mathbb{R}^3$, which lies in the compact convex set $K = \operatorname{conv} \Phi(P)$. The bound is $\mu f_N t^\star$, where $(c, t^\star)$ is the highest point of K above c. This point lies on a supporting plane H of K, and $K \cap H = \operatorname{conv}(\Phi(P) \cap H)$, so Carathéodory's theorem in the plane H writes it as a convex combination $\sum_j \beta_j \Phi(x_j)$ of $s \leq 3$ points. The measure $f_N \sum_j \beta_j \delta_{x_j}$ is compatible and attains the bound. □

Corollary 4.5 (Polygonal contact patch). *Suppose additionally that P is a convex polygon with vertices $v_1, \dots, v_K$.*

(i) The coarse upper bound is the linear program

$$\boxed{\begin{aligned} \overline{h}(q; y_0, \mathcal{M}_0) = \mu \max_{\lambda_1,\dots,\lambda_K \geq 0} \quad & \sum_{k=1}^{K} \lambda_k \|a + bJv_k\|_2 \\ s.t. \quad & \sum_{k=1}^{K} \lambda_k = f_N, \qquad \sum_{k=1}^{K} \lambda_k v_k = m_1, \end{aligned}} \tag{34}$$

which has an optimal solution with at most three nonzero weights.

(ii) For a vertex triangle $\Delta = \operatorname{conv}\{v_i, v_j, v_k\}$ containing c, let $\lambda^{\Delta}(y_0) \in \mathbb{R}^3_+$ be the unique weights with $\sum \lambda^{\Delta} = f_N$, $\sum \lambda^{\Delta}_l v_l = m_1$ (homogeneous barycentric coordinates, linear in y_0). Then

$$\overline{h}(q; y_0, \mathcal{M}_0) = \mu \max_{\Delta \ni c} \sum_{l \in \Delta} \lambda^{\Delta}_l(y_0) \, \|a + bJv_l\|_2, \tag{35}$$

the maximum over the vertex triangles containing c. If P is a triangle, $\overline{h}(q; \cdot, \mathcal{M}_0)$ is linear in y_0. If P is a convex quadrilateral (e.g. a rectangle), each of its two diagonal triangulations has a triangle containing c, and the bound is the larger of the corresponding values.

Proof. (i) By Theorem 4.4, an optimal measure is $\sum_{j=1}^{s} \lambda_j \delta_{x_j}$ with $s \leq 3$. Write each x_j as a convex combination $\sum_k \alpha_{jk} v_k$ of the vertices. Moving the mass $\lambda_j \alpha_{jk}$ from x_j to v_k preserves the zeroth and first moments and, by convexity of $g(\cdot; q)$ (Lemma 3.13(i)), does not decrease the objective. This is the reduction to the extreme points of P in Yann's Proposition 4.4. Hence an optimal measure supported on the vertices exists, which gives the LP. The LP has three equality constraints, so it has a basic optimal solution with at most three nonzero weights. (ii) The basic feasible solutions are supported on affinely independent vertices whose convex hull contains c; completing them to a vertex triangle containing c (with zero weights) gives the weights $\lambda^{\Delta}(y_0)$, which are unique since the vertices of Δ are affinely independent. An LP attains its maximum at a basic solution. The diagonal triangulations of a quadrilateral cover it. □

Remark 4.6 (Connection with Yann's formulation). *Theorem 4.4 and Corollary 4.5 are the Coulomb case $C = B(0, \mu)$ of Proposition 4.4 in Yann's draft, with the same reference point. With his notation $x^{\perp} = -Jx$, his moment $m_N = -\int \langle \rho_T, x^{\perp} \rangle$ is our τ_N, his twist (ω_N, v_T) is our $q = (v_T, \omega_N)$, and his integrand is $s_C(\omega_N x^{\perp} - v_T) = \mu \|v_T + \omega_N Jx\|_2 = \mu \, g(x; q)$. His set $f_N \, D(c^{\perp})$ is $\mathcal{X}^+(y_0, \mathcal{M}_0)$ (components reordered), so that $f_N \, s_{D(c^{\perp})}(-[\omega_N, v_T]) = \overline{h}(q; y_0, \mathcal{M}_0)$, and the concavity of $a \mapsto s_{D(a)}$ is Proposition 3.11(iii). His special cases are the CoP at a vertex of P (a single point load, both bounds equal $\mu f_N g(c; q)$) and $\omega_N = 0$ (both bounds equal $\mu f_N \|v_T\|_2$). As he notes, $D(\cdot)$ is the place to add priors on the distribution; the taxel measurements of the next sections are such priors.*

The two bounds differ in general, which shows that the coarse measurement does not determine the admissible wrench set.

Example 4.7 (Segment patch). *Let $P = [0, L] \times \{0\}$, $c = \left(\frac{L}{2}, 0\right)$, $\nu_N^{(1)} = f_N \delta_c$ and $\nu_N^{(2)} = \frac{f_N}{2}\delta_{(0,0)} + \frac{f_N}{2}\delta_{(L,0)}$. Both have $\mathcal{M}_0(\nu_N^{(1)}) = \mathcal{M}_0(\nu_N^{(2)}) = \left(f_N, f_N \left(\frac{L}{2}, 0\right)\right)$. For the unit rotation about the CoP, $q_c = (-Jc, 1)$, so that $g(x; q_c) = \|x - c\|_2$,*

$$h_{\mathcal{X}(\nu_N^{(1)})}(q_c) = 0 \quad \neq \quad h_{\mathcal{X}(\nu_N^{(2)})}(q_c) = \mu f_N \frac{L}{2} = \overline{h}(q_c; y_0, \mathcal{M}_0). \tag{36}$$

The results of this section are recovered in Section 7 as the special case of a single taxel $P_1 = P$.

5 Taxel grids: setting and common lemmas

We now assume that a tactile array measures the normal force carried by each taxel, with or without the coarse measurement. Tactile arrays are grids of rectangular taxels, and we restrict to them; other layouts are discussed in Remark 7.4. This section fixes the setting and collects the lemmas used in Sections 6 and 7.

Definition 5.1 (Taxel grid)**.** *Let $P = [0, L] \times [0, \ell]$ be the sensor area. For $k_x, k_y \geq 1$, the grid $G(k_x, k_y)$ divides P into $r = k_x k_y$ taxels $P_1, \dots, P_r$: the rectangles $[\mathrm{lo}_i, \mathrm{hi}_i)$ (componentwise half-open) of size $w_x \times w_y = \frac{L}{k_x} \times \frac{\ell}{k_y}$, those of the last column and row being closed on their outer side, so that every point of P belongs to exactly one taxel. We write $\overline{P_i} = [\mathrm{lo}_i, \mathrm{hi}_i]$ for the closed taxel, m_i for its center, $v_{i1}, \dots, v_{i4}$ for its corners, and $D = \sqrt{w_x^2 + w_y^2}$ for the taxel diameter. A load exactly on an edge shared by two taxels is assigned to one of them by this convention; a real sensor cannot resolve it, but a pressure field with a density puts no force on edges, and Lemma 5.5 shows that the convention does not change the bounds.*

Definition 5.2 (Taxel force measurement)**.** *Given the true unknown normal force measure $\nu_N^\star$ and a taxel grid, the taxel force measurement is*

$$y_F = \mathcal{M}_F(\nu_N^\star) \triangleq (\nu_N^\star(P_1), \dots, \nu_N^\star(P_r)), \tag{37}$$

with measurement functions $\phi_i = \mathbf{1}_{P_i}$, $i = 1, \dots, r$. The measured taxel forces are denoted $F_i \triangleq \nu_N^\star(P_i)$, and $f_N = \sum_i F_i$. Since $\sum_i \mathbf{1}_{P_i} = 1$, $\mathcal{M}_F$ satisfies Assumption 3.9. For the grid $G(k_x, k_y)$ we write $\mathcal{M}_F(k_x, k_y)$; the dyadic grids $\mathcal{M}_F(2^{p_x}, 2^{p_y})$ are Yann's family $\mathcal{M}(T; p_x, p_y)$.

Definition 5.3 (Combined coarse and taxel force measurement)**.** *The combined measurement is the concatenation $\mathcal{M}_T \triangleq \mathcal{M}_0 \vee \mathcal{M}_F$,*

$$y_T = \mathcal{M}_T(\nu_N^\star) = (f_N, m_1, F_1, \dots, F_r), \qquad \Pi_{\mathcal{M}_T}(y_T) = \Pi_{\mathcal{M}_0}(y_0) \cap \Pi_{\mathcal{M}_F}(y_F). \tag{38}$$

We write $\mathcal{M}_T(k_x, k_y) \triangleq \mathcal{M}_0 \vee \mathcal{M}_F(k_x, k_y)$ (Yann's family $\mathcal{M}(T_c; p_x, p_y)$ for dyadic grids).

Proposition 5.4 (Information chains)**.** *Let $\ell > 0$.*

(i) $\mathcal{M}_F(k_x, k_y) \succeq \mathcal{M}_F(k'_x, k'_y)$ if and only if k'_x divides k_x and k'_y divides k_y, and the same holds for $\mathcal{M}_T$. In particular, the dyadic grids form chains, and 3×3 and 4×4 grids are not comparable. $\mathcal{M}_F(1, 1)$ measures the resultant normal force alone, and $\mathcal{M}_T(1, 1)$ is equivalent to $\mathcal{M}_0$ (each $\succeq$ the other).

(ii) $\mathcal{M}_T \succeq \mathcal{M}_0$ and $\mathcal{M}_T \succeq \mathcal{M}_F$. If $r \geq 2$, $\mathcal{M}_0$ and $\mathcal{M}_F$ are not comparable.

Proof. By Proposition 3.15(ii). (i) If $k'_x \mid k_x$ and $k'_y \mid k_y$, every coarse taxel is a union of fine taxels (the half-open conventions agree), so $\mathbf{1}_{P'_j} = \sum_{i : P_i \subseteq P'_j} \mathbf{1}_{P_i}$ pointwise; concatenation with $\mathcal{M}_0$ preserves the order. Conversely, if $k'_x \nmid k_x$, some coarse breakpoint jL/k'_x lies strictly inside a fine column. Along a horizontal segment that crosses it inside one fine taxel, the indicator of a coarse taxel jumps, while every fine indicator is constant and $1, x_1, x_2$ are continuous: it is not a combination of them. The same holds in y. (ii) Proposition 3.15(iii); a taxel indicator is not affine on P, and x_1 is not constant on a taxel. □

Loads on taxel boundaries. The finite-dimensional formulas below are attained by point loads that may sit on an edge or a corner owned by a neighboring taxel. We call $\nu_N = \sum_i \nu_i$ a *closed-cell measure* for the data $(F_1, \dots, F_r)$ if each $\nu_i \in \mathcal{M}_+(P)$ is carried by the closed taxel $\overline{P_i}$ and $\nu_i(\overline{P_i}) = F_i$. Every compatible measure is one (take ν_i the restriction of ν_N to P_i), and the converse holds up to an arbitrarily small move:

Lemma 5.5 (Boundary loads). *Let $\mathcal{M} \in \{\mathcal{M}_F, \mathcal{M}_T\}$ on a taxel grid, $y = \mathcal{M}(\nu_N^\star)$, and let $\nu_N = \sum_i \nu_i$ be a closed-cell measure for the taxel forces of y (with $m_1(\nu_N) = m_1$ for $\mathcal{M}_T$). For every $\varepsilon \in (0, 1]$ there is $\nu_N^\varepsilon \in \Pi_{\mathcal{M}}(y)$ with $|h_{\mathcal{X}(\nu_N^\varepsilon)}(q) - h_{\mathcal{X}(\nu_N)}(q)| \le \varepsilon\mu|b| \sum_i F_i \operatorname{diam}(P_i)$ for every $q = (a, b)$. Consequently, the infimum and the supremum of $h_{\mathcal{X}(\nu_N)}(q)$ over $\Pi_{\mathcal{M}}(y)$ equal those over the closed-cell measures.*

Proof. For each i with $F_i > 0$, let p_i be the local CoP of $\nu_N^\star$ in P_i; it lies in the half-open rectangle P_i, since an average of points of $[\alpha, \beta)$ lies in $[\alpha, \beta)$. Let ν_i^ε be the image of ν_i under $x \mapsto (1-\varepsilon)x + \varepsilon p_i$, which maps $\overline{P_i}$ into P_i coordinate-wise. Then $\nu_N^\varepsilon = \sum_i \nu_i^\varepsilon$ has taxel forces F_i, and first moment $(1-\varepsilon)m_1(\nu_N) + \varepsilon \sum_i F_i p_i = m_1(\nu_N)$ when $m_1(\nu_N) = m_1(\nu_N^\star)$, so $\nu_N^\varepsilon \in \Pi_{\mathcal{M}}(y)$. Each mass moves by at most $\varepsilon \operatorname{diam}(P_i)$, which gives the estimate by Lemma 3.13(ii). □

Lemma 5.6 (Concentration and spreading). *Let $\nu_N = \sum_i \nu_i$ be a closed-cell measure, with local first moments $u_i \triangleq \int x \, d\nu_i(x)$.*

(i) Concentration. For $F_i > 0$, $c_i \triangleq u_i/F_i \in \overline{P_i}$, and the closed-cell measure $\nu_N^c \triangleq \sum_{i:F_i>0} F_i \delta_{c_i}$ has the first moment of ν_N and

$$\mathcal{X}(\nu_N^c) \subseteq \mathcal{X}(\nu_N). \tag{39}$$

If ν_N is compatible (ν_i its restriction to P_i), then $c_i \in P_i$ and ν_N^c is compatible.

(ii) Spreading. There are corner weights $\lambda_{ik} \ge 0$ with $\sum_k \lambda_{ik} = F_i$ and $\sum_{i,k} \lambda_{ik} v_{ik} = m_1(\nu_N)$ such that the corner-supported measure $\nu_N^s \triangleq \sum_{i,k} \lambda_{ik} \delta_{v_{ik}}$ satisfies

$$\mathcal{X}(\nu_N) \subseteq \mathcal{X}(\nu_N^s). \tag{40}$$

In words: concentrating the force of each taxel at its local CoP shrinks the wrench set, and spreading it to the taxel corners enlarges it, both without changing the taxel forces or the first moment.

Proof. We use the description of $\mathcal{X}(\nu_N)$ by tangential measures ν_T with $\|\nu_T(E)\|_2 \le \mu\nu_N(E)$ (footnote to (13)). (i) c_i is an average of points of $\overline{P_i}$ (of P_i if ν_N is compatible). A wrench of ν_N^c is given by vectors $\gamma_i \in B(0, \mu)$ at the atoms: $(\sum_i F_i\gamma_i, \sum_i c_i \times F_i\gamma_i)$. The tangential measure $\nu_T = \sum_i \gamma_i \nu_i$ is admissible for ν_N, with resultant force $\sum_i F_i\gamma_i$ and moment $\sum_i u_i \times \gamma_i$, the same wrench. (ii) On $\overline{P_i}$, let $t = (x_1 - \mathrm{lo}_{i,1})/w_x$ and $s = (x_2 - \mathrm{lo}_{i,2})/w_y$, and let $\alpha_{ik}(x)$ be the bilinear weights $(1-t)(1-s)$, $t(1-s)$, ts, $(1-t)s$ of the corners, so that $\alpha_{ik} \ge 0$, $\sum_k \alpha_{ik} = 1$ and $\sum_k \alpha_{ik}(x) v_{ik} = x$. Let $\lambda_{ik} = \int \alpha_{ik} \, d\nu_i$, which has the stated sums. A wrench of ν_N is given by an admissible field γ_T; set $\rho_{ik} = \int \alpha_{ik}\gamma_T \, d\nu_i$. Then $\|\rho_{ik}\|_2 \le \mu\lambda_{ik}$, $\sum_{i,k} \rho_{ik} = f_T$ and $\sum_{i,k} v_{ik} \times \rho_{ik} = \sum_i \int (\sum_k \alpha_{ik}(x) v_{ik}) \times \gamma_T(x) \, d\nu_i(x) = \tau_N$, so the corner forces ρ_{ik} realize the same wrench for ν_N^s. □

Spreading is the reduction to the extreme points of the patch in Yann's draft (his Eq. (33) and Proposition 4.4), applied taxel by taxel and at the level of sets.

Remark 5.7 (Intuition). *The infinite-dimensional problems over measures reduce to finite-dimensional ones through Lemma 5.6 (Figure 2).*

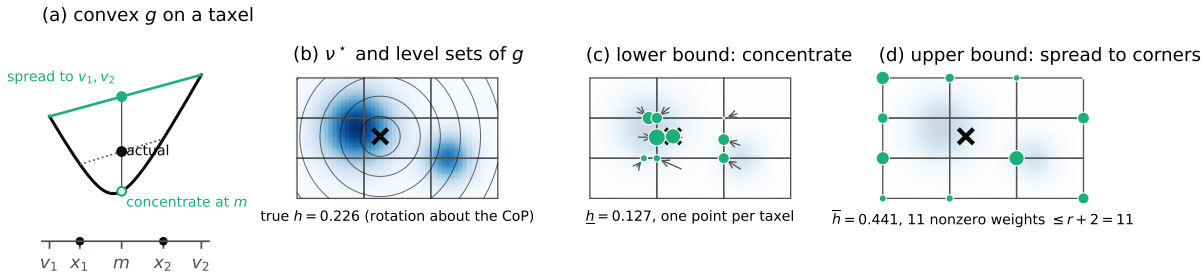


Figure 2: Concentrate vs. spread (Remark 5.7), 3×3 taxels, unit rotation about the CoP $q_c = (-Jc, 1)$, so $g(x; q_c) = \|x - c\|_2$. (a) For masses at x_1, x_2 with CoP m: concentrating at m lowers the objective (Jensen), spreading to the vertices v_1, v_2 raises it (chord). (b) True measure and level sets of g. (c) Minimizer of Theorem 7.1: one Dirac per taxel, pulled from the local CoPs of the taxels toward the CoP (arrows). (d) Maximizer of Theorem 7.2: weights on the taxel corners, pushed away from the CoP, with at most $r + 2$ nonzero weights. Marker areas are proportional to force.

- *Lower bounds = "concentrate". One point load per taxel, at its local CoP, is the least favorable distribution with given taxel forces and first moment: every wrench it can carry is carried by the original measure. Needed for all lower bounds: $\mathcal{M}_0$ (Theorem 4.2, a single point at the CoP), $\mathcal{M}_F$ (Theorem 6.1) and $\mathcal{M}_T$ (Theorem 7.1).*
- *Upper bounds = "spread to corners". Splitting each mass among the corners of its taxel, with weights whose average position is the mass's position, keeps the taxel forces and the first moment, and every wrench of the original measure remains available. Needed for all upper bounds: $\mathcal{M}_0$ (Corollary 4.5, corners of P), $\mathcal{M}_F$ (Theorem 6.1) and $\mathcal{M}_T$ (Theorem 7.2).*
- *Boundaries. A corner of a taxel may belong to its neighbor; by Lemma 5.5, corner-supported measures nevertheless give the exact values.*

Remark 5.8 (Compatible CoPs)**.** *The first moments of the closed-cell measures for y_F form the Minkowski sum $\sum_i F_i \overline{P_i}$, by Lemma 5.6(i) and its converse ($\sum_i F_i \delta_{c_i}$ is a closed-cell measure for any $c_i \in \overline{P_i}$). Since the closed taxels are translates $m_i + R_0$ of one rectangle $R_0 = [-\frac{w_x}{2}, \frac{w_x}{2}] \times [-\frac{w_y}{2}, \frac{w_y}{2}]$, this is, for $f_N > 0$ and in terms of the CoP,*

$$\sum_{i=1}^{r} \frac{F_i}{f_N} \overline{P_i} = \bar{c}_F + R_0, \qquad \bar{c}_F \triangleq \frac{1}{f_N} \sum_{i=1}^{r} F_i m_i : \tag{41}$$

a taxel-shaped rectangle centered at the force-weighted average of the taxel centers (Yann's observation), of diameter D whatever the forces. The compatible CoPs are dense in it. The bounds do not use this set; it is shown in the figures, and an observation y_T is consistent only if its CoP lies in $\bar{c}_F + R_0$.

6 $\mathcal{M}_F$ case (taxel forces)

Known: the taxel forces F_i. **Unknown:** the distribution of $\nu_N^\star$ inside each taxel, hence also its CoP. Without the first-moment constraint, the taxels decouple and both bounds have closed forms.

Theorem 6.1 (Exact bounds for taxel force measurements)**.** *On a taxel grid, for every* $q = (a, b) \in \mathbb{R}^2 \times \mathbb{R}$,

$$\boxed{\underline{h}(q; y_F, \mathcal{M}_F) = \mu \sum_{i=1}^{r} F_i \min_{x \in \overline{P_i}} \|a + bJx\|_2, \qquad \overline{h}(q; y_F, \mathcal{M}_F) = \mu \sum_{i=1}^{r} F_i \max_{k=1,\dots,4} \|a + bJv_{ik}\|_2.} \tag{42}$$

For $b \neq 0$*, with the instantaneous center of rotation* $p = Ja/b$ *and* $\Delta_i = m_i - p$*, these are* $\mu|b| \sum_i F_i\, d_i^{\min}$ *and* $\mu|b| \sum_i F_i\, d_i^{\max}$*, with*

$$d_i^{\min} = \min_{x \in \overline{P_i}} \|x - p\|_2 = \left\| \left(\left(|\Delta_{i,1}| - \tfrac{w_x}{2} \right)_+, \left(|\Delta_{i,2}| - \tfrac{w_y}{2} \right)_+ \right) \right\|_2, \tag{43}$$

$$d_i^{\max} = \max_{x \in \overline{P_i}} \|x - p\|_2 = \left\| \left(|\Delta_{i,1}| + \tfrac{w_x}{2}, |\Delta_{i,2}| + \tfrac{w_y}{2} \right) \right\|_2. \tag{44}$$

For $b = 0$ *both bounds equal* $\mu f_N \|a\|_2$.

Proof. By Lemma 5.5, we may optimize over the closed-cell measures, a product of the families $\{\nu_i \text{ on } \overline{P_i},\ \nu_i(\overline{P_i}) = F_i\}$, and the objective $\mu \sum_i \int g\, d\nu_i$ is a sum of one term per family. For each term, $F_i \min_{\overline{P_i}} g \le \int g\, d\nu_i \le F_i \max_{\overline{P_i}} g$, with equality for a point load at a minimizer or a maximizer; the maximum of the convex function $g(\cdot; q)$ over a rectangle is attained at a corner. The closed forms are the distance from p to the rectangle and to its farthest corner. □

Remark 6.2 (Relation with the earlier formulation)**.** *The bounds* (42) *are the "cellwise" bounds of the combined measurement (the* $\mathcal{M}_T$ *problem without its first-moment constraint), which are exact for* $\mathcal{M}_F$*. An earlier version of this note took the torque about the CoP of each candidate measure. Under* $\mathcal{M}_F$ *the CoP is unknown, and that formulation leads to a bilevel, generally nonconvex problem (an outer optimization over the compatible CoPs of Remark 5.8). Its values are valid bounds on the support of the true set about the true CoP, but they compare twists written about different points for different candidates (Remark 3.5), and cannot be used to test a load given in a fixed frame. They are not used.*

7 $\mathcal{M}_T$ case (taxel forces and CoP)

Known: the taxel forces F_i and the first moment m_1, i.e., the CoP. **Unknown:** the distribution of $\nu_N^\star$ inside each taxel. The taxel constraints act on each taxel separately, but the first-moment constraint $\int_P x\, d\nu_N = m_1$ couples the taxels.

Theorem 7.1 (Exact lower bound for combined measurements)**.** *On a taxel grid, given* $y_T = (f_N, m_1, F_1, \dots, F_r) = \mathcal{M}_T(\nu_N^\star)$*, for every* $q = (a, b)$,

$$\boxed{\begin{aligned} \underline{h}(q; y_T, \mathcal{M}_T) = \mu \min_{u_1,\dots,u_r \in \mathbb{R}^2} \quad & \sum_{i=1}^{r} \|F_i a + bJu_i\|_2 \\ \text{subject to} \quad & F_i\, \mathrm{lo}_i \le u_i \le F_i\, \mathrm{hi}_i, \qquad i = 1, \dots, r, \\ & \sum_{i=1}^{r} u_i = m_1. \end{aligned}} \tag{45}$$

The variables are the local first moments (inequalities componentwise); for $F_i > 0$, $u_i = F_i c_i$ *with* $c_i \in \overline{P_i}$ *the local CoP, and the objective is* $\mu \sum_i F_i\, g(c_i; q)$*, while* $F_i = 0$ *forces* $u_i = 0$*. The problem is an SOCP.*

Proof. ($\geq$) For a compatible ν_N, the local first moments $u_i = \int_{P_i} x\, d\nu_N$ are feasible, and $\int_{P_i} g\, d\nu_N \geq \|F_i a + bJu_i\|_2$ by Lemma 3.13(iii) applied to the restriction of ν_N to P_i. ($\leq$) For feasible u_i, the concentrated measure $\sum_{i:F_i>0} F_i \delta_{u_i/F_i}$ is a closed-cell measure with first moment m_1 that attains the objective; by Lemma 5.5, compatible measures come arbitrarily close. The feasible set is compact and nonempty (it contains the local first moments of $\nu_N^\star$), and the objective is convex and continuous; epigraph variables give an SOCP. □

Theorem 7.2 (Exact upper bound for combined measurements)**.** *Under the assumptions of Theorem 7.1, for every $q = (a, b)$,*

$$\begin{aligned} \overline{h}(q; y_T, \mathcal{M}_T) = \mu \max_{\lambda_{ik} \geq 0} \quad & \sum_{i=1}^{r} \sum_{k=1}^{4} \lambda_{ik} \, \|a + bJv_{ik}\|_2 \\ \text{subject to} \quad & \sum_{k=1}^{4} \lambda_{ik} = F_i, \qquad i = 1, \ldots, r, \\ & \sum_{i=1}^{r} \sum_{k=1}^{4} \lambda_{ik} v_{ik} = m_1. \end{aligned} \tag{46}$$

This is a linear program, and an optimal solution exists with at most $r + 2$ nonzero weights.

Proof. ($\leq$) For a compatible ν_N, Lemma 5.6(ii) gives a feasible λ with $h_{\mathcal{X}(\nu_N)}(q) \leq h_{\mathcal{X}(\nu_N^s)}(q)$, which is the objective of λ. ($\geq$) For a feasible λ, the corner-supported measure $\sum_{i,k} \lambda_{ik} \delta_{v_{ik}}$ is a closed-cell measure with first moment m_1 that attains the objective; by Lemma 5.5, compatible measures come arbitrarily close. The LP is feasible (by spreading $\nu_N^\star$) and bounded, and a basic optimal solution has at most as many nonzero entries as equality constraints, i.e., $r + 2$. □

Remark 7.3 (Recovering the coarse and taxel-force cases)**.** *For a single taxel ($k_x = k_y = 1$), the constraint of (45) forces $u_1 = m_1$, and Theorem 7.1 reduces to Theorem 4.2; Theorem 7.2 reduces to Corollary 4.5 for the rectangle P, with $r + 2 = 3$ nonzero weights. Dropping the first-moment constraint in (45) and (46) decouples the taxels and gives the $\mathcal{M}_F$ bounds of Theorem 6.1. Consistently with Theorem 3.16 and $\mathcal{M}_T \succeq \mathcal{M}_F$, the $\mathcal{M}_T$ bounds lie between the $\mathcal{M}_F$ bounds:*

$$\underline{h}(q; y_F, \mathcal{M}_F) \leq \underline{h}(q; y_T, \mathcal{M}_T) \leq \overline{h}(q; y_T, \mathcal{M}_T) \leq \overline{h}(q; y_F, \mathcal{M}_F). \tag{47}$$

The difference is the value of measuring the CoP in addition to the taxel forces.

Remark 7.4 (Other taxel layouts)**.** *The restriction to uniform rectangular grids only simplifies the statements.*

- *Non-uniform rectangular grids and convex polygonal cells (e.g. hexagonal arrays): Lemma 5.6 holds with barycentric coordinates of the cell (after a triangulation) instead of bilinear ones. Theorems 6.1, 7.1 and 7.2 hold with the minimum over the cell, the maximum over its vertices, the constraint $u_i \in F_i \overline{P_i}$ (linear for a polygon) and the LP over all vertices, again with at most $r + 2$ nonzero weights.*
- *Boundaries: Lemma 5.5 needs that each closed cell can be pushed into its own cell toward an interior point, which holds for half-open rectangles. Otherwise the formulas are the exact bounds of the closed-cell measures, which are valid and possibly conservative for the actual cells by Theorem 3.16.*
- *Refinement: a partition that refines another is more informative (Proposition 3.15(ii)). Theorem 8.4 and the rectangle of Remark 5.8 use uniform rectangular cells.*

8 Comparison of measurements and tightening under refinement

All results of this section follow from Theorem 3.16, the information chains of Proposition 5.4, and the closed forms of Theorem 6.1.

Proposition 8.1 (Comparison of coarse, taxel and combined measurements)**.** *For every twist* $q = (a, b) \in \mathbb{R}^2 \times \mathbb{R}$,

$$\boxed{\begin{aligned}\max\{\underline{h}(q; y_0, \mathcal{M}_0),\, \underline{h}(q; y_F, \mathcal{M}_F)\} \le \underline{h}(q; y_T, \mathcal{M}_T) \le h_{\mathcal{X}(\nu_N^\star)}(q),\\ h_{\mathcal{X}(\nu_N^\star)}(q) \le \overline{h}(q; y_T, \mathcal{M}_T) \le \min\left\{\overline{h}(q; y_0, \mathcal{M}_0),\, \overline{h}(q; y_F, \mathcal{M}_F)\right\},\end{aligned}} \tag{48}$$

with $\underline{h}(q; y_0, \mathcal{M}_0) = \mu\|f_N a + bJm_1\|_2$*. The bounds of* $\mathcal{M}_0$ *and* $\mathcal{M}_F$ *are not ordered in general, in either direction.*

Proof. $\mathcal{M}_T = \mathcal{M}_0 \vee \mathcal{M}_F$, Proposition 3.15(iii) and Theorem 3.16. For the last claim: with a single taxel, $\mathcal{M}_F$ measures only f_N, and its bounds are wider than those of $\mathcal{M}_0$ (e.g. $\underline{h}(q; y_F, \mathcal{M}_F) = \mu f_N |b| \operatorname{dist}(p, P) < \underline{h}(q; y_0, \mathcal{M}_0) = \mu f_N |b| \|c - p\|_2$ when the CoP c is not the point of P closest to p). With fine taxels, the $\mathcal{M}_F$ bounds converge to the truth (Proposition 8.3), while the $\mathcal{M}_0$ bounds do not in general (Example 4.7). □

Proposition 8.2 (Tightening under taxel refinement)**.** *Let* $\mathcal{M}'_F$, $\mathcal{M}'_T$ *and* $\mathcal{M}_F$, $\mathcal{M}_T$ *be the taxel-force and combined measurements of the grids* $G(k'_x, k'_y)$ *and* $G(k_x, k_y)$*, with* $k_x \mid k'_x$ *and* $k_y \mid k'_y$ *(e.g. dyadic grids, each doubling the previous one). Then, for every* q *and* $\mathcal{M} \in \{\mathcal{M}_F, \mathcal{M}_T\}$*, with* $y = \mathcal{M}(\nu_N^\star)$ *and* $y' = \mathcal{M}'(\nu_N^\star)$*,*

$$\boxed{\underline{h}(q; y, \mathcal{M}) \le \underline{h}(q; y', \mathcal{M}') \le h_{\mathcal{X}(\nu_N^\star)}(q) \le \overline{h}(q; y', \mathcal{M}') \le \overline{h}(q; y, \mathcal{M}),} \tag{49}$$

and the guaranteed sets grow and the possible sets shrink.

Proof. Proposition 5.4(i) and Theorem 3.16. □

The monotonicity is associated with nested grids, not with the number of taxels r: a 3×3 and a 4×4 grid carry different information (Proposition 5.4(i)), and nothing guarantees that the second gives tighter bounds. The gap bounds below, on the other hand, only require small taxel diameters.

Proposition 8.3 (Resolution-dependent gap bounds)**.** *For every* $q = (a, b) \in \mathbb{R}^2 \times \mathbb{R}$ *and* $\mathcal{M} \in \{\mathcal{M}_F, \mathcal{M}_T\}$*,*

$$\boxed{\begin{aligned}0 \le \overline{h}(q; y, \mathcal{M}) - \underline{h}(q; y, \mathcal{M}) &\le \mu \sum_{i=1}^{r} F_i \left(\max_{\overline{P_i}} g(\cdot; q) - \min_{\overline{P_i}} g(\cdot; q)\right)\\ &\le \mu|b| \sum_{i=1}^{r} F_i \operatorname{diam}(P_i) \le \mu|b| f_N \max_i \operatorname{diam}(P_i),\end{aligned}} \tag{50}$$

with equality in the first inequality for $\mathcal{M}_F$*. Hence, along any sequence of taxel grids with taxel diameter* $D \to 0$*, the four bounds of* $\mathcal{M}_F$ *and* $\mathcal{M}_T$ *converge to* $h_{\mathcal{X}(\nu_N^\star)}(q)$*.*

Proof. For $\mathcal{M}_F$ the gap is the difference of the two closed forms of Theorem 6.1; for $\mathcal{M}_T$ it is smaller, by (47). By Lemma 3.13(ii), $\max_{\overline{P_i}} g - \min_{\overline{P_i}} g \le |b| \operatorname{diam}(P_i)$. Convergence follows since $h_{\mathcal{X}(\nu_N^\star)}(q)$ lies between the bounds. □

With the torque taken about the CoP of each candidate measure, the corresponding bound for $\mathcal{M}_F$ had an additional factor 2, which came from the unknown reference point.

The next result answers how much one refinement step tightens the bounds. For a twist with $b \neq 0$, let $p = Ja/b$ be its instantaneous center of rotation and $u(x) = (x-p)/\|x-p\|_2$ the unit radial direction at $x \neq p$.

Theorem 8.4 (First-order rate of tightening)**.** *Consider a taxel grid $G(k_x, k_y)$ (Definition 5.1), with taxels of size $w_x \times w_y$ and diameter $D = \sqrt{w_x^2 + w_y^2}$, and let $\nu_N^\star$ have a density $\rho_N \leq \bar{\rho}$ with respect to area. Then, for every twist $q = (a, b)$ with $b \neq 0$,*

$$\left| \overline{h}(q; y_F, \mathcal{M}_F) - \underline{h}(q; y_F, \mathcal{M}_F) - \mu|b| \int_P \left(|u_1(x)|\, w_x + |u_2(x)|\, w_y\right) d\nu_N^\star(x) \right| \leq \tag{51}$$

$$\mu|b|\, \bar{\rho}\, D^2 \left(\tfrac{15}{4} \sqrt{\pi |P|} + \tfrac{9\pi}{4} D \right), \tag{52}$$

where $|P|$ is the area of P. In particular, for square cells of side h,

$$\overline{h}(q; y_F, \mathcal{M}_F) - \underline{h}(q; y_F, \mathcal{M}_F) = \mu|b|\, C_\nu(p)\, h + O(h^2), \tag{53}$$

$$C_\nu(p) \triangleq \int_P \left(|u_1| + |u_2|\right) d\nu_N^\star \in \left[f_N, \sqrt{2} f_N \right], \tag{54}$$

so that, if $f_N > 0$, doubling k_x and k_y divides the $\mathcal{M}_F$ gap by $2 + O(h)$.

Proof. By Theorem 6.1, the gap is $\mu|b| \sum_C F_C e_C$, where the sum is over the cells, $F_C = \nu_N^\star(C)$, and $e_C = \max_{\overline{C}} d - \min_{\overline{C}} d$ with $d(x) = \|x - p\|_2$. Let m be the center of a cell C, $z = m - p$, and $\ell_C \triangleq |u_1(m)| w_x + |u_2(m)| w_y$, which is the range of the linear function $x \mapsto \langle u(m), x \rangle$ over $\overline{C}$.

Far cells ($\|z\|_2 \geq D$). For $x \in \overline{C}$, write $y = x - p$ and $d(x) = \langle u(m), y \rangle + r(x)$. Since the component of y orthogonal to $u(m)$ is that of $x - m$, $0 \leq r(x) = \frac{\|y\|_2^2 - \langle u(m), y \rangle^2}{\|y\|_2 + \langle u(m), y \rangle} \leq \frac{(D/2)^2}{\|z\|_2}$, because $\|x - m\|_2 \leq D/2$ and $\|y\|_2, \langle u(m), y \rangle \geq \|z\|_2 - D/2 \geq \|z\|_2/2$. The maximum and the minimum of d over $\overline{C}$ are therefore within $D^2/(4\|z\|_2)$ of those of the linear part, so $|e_C - \ell_C| \leq D^2/(4\|z\|_2)$. Moreover, for $x \in C$, $\|u(x) - u(m)\|_2 \leq 2\|x - m\|_2/\|z\|_2 \leq D/\|z\|_2$, so $|\ell_C - \ell(x)| \leq D^2/\|z\|_2$, with $\ell(x) \triangleq |u_1(x)| w_x + |u_2(x)| w_y$, and $\|z\|_2 \geq \frac{2}{3}\|x - p\|_2$. Hence $|F_C e_C - \int_C \ell\, d\nu_N^\star| \leq \frac{15}{8} D^2 \int_C \frac{d\nu_N^\star(x)}{\|x-p\|_2}$.

Near cells ($\|z\|_2 < D$). Both e_C and $\ell(x)$ lie in $[0, D]$ (Lemma 3.13(ii) and Cauchy–Schwarz), so $|F_C e_C - \int_C \ell\, d\nu_N^\star| \leq D F_C$. These cells lie in the disk $B(p, \frac{3}{2}D)$, which carries at most $\bar{\rho}\, \frac{9\pi}{4} D^2$.

Summing, the left-hand side of (51) is at most $\mu|b| \left(\frac{15}{8} D^2 \bar{\rho} \int_P \frac{dA}{\|x-p\|_2} + \frac{9\pi}{4} \bar{\rho} D^3 \right)$, and $\int_P \frac{dA}{\|x-p\|_2} \leq 2\sqrt{\pi |P|}$, the value for the disk of area $|P|$ centered at p. For square cells, $\ell(x) = h(|u_1| + |u_2|)$ and $1 \leq |u_1| + |u_2| \leq \sqrt{2}$ for a unit vector. The ratio of two consecutive gaps is $\frac{C_\nu(p) h/2 + O(h^2)}{C_\nu(p) h + O(h^2)}$ with $C_\nu(p) \geq f_N > 0$. □

Remark 8.5 (Reading the rate)**.**

- *In terms of the diameter $D = \sqrt{2} h$ of square cells, the $\mathcal{M}_F$ gap is $\frac{C_\nu(p)}{\sqrt{2} f_N}\, \mu|b| f_N D + O(D^2)$, between $\frac{1}{\sqrt{2}}$ and 1 times the bound of Proposition 8.3: that bound is sharp up to a factor at most $\sqrt{2}$. The constant $C_\nu(p)$ depends on how the load is distributed around the center of rotation, and is largest when the load is seen from p along the diagonals of the grid.*
- *Refining in one direction only, $G(k_x, k_y) \to G(2k_x, k_y)$, halves the term $\int |u_1|\, d\nu\, w_x$ only: the gain depends on the direction of the load seen from p.*
- *The bounded density excludes point loads; for measures with concentrated loads near the cell boundaries, one refinement step can leave the gap almost unchanged at coarse resolution.*
- *The same first-order term bounds the $\mathcal{M}_T$ gap from above, by (47). How much the first-moment constraint reduces the gap at high resolution, i.e., the value of measuring the CoP in addition to the taxel forces, is left open here and quantified numerically in the experimental report.*

9 Certified load factor

A controller does not need the bounds direction by direction. It needs to know whether a given load w can be carried, whether it certainly cannot, and by how much the contact should be strengthened (e.g., by squeezing harder). This is a question about the ray $\{sw : s \geq 0\}$: how far along it the guaranteed and the possible sets extend. This section shows that both answers are convex programs for the taxel measurements, without constructing the sets.

Reference point of the load. The load acting on a grasped object is known about a point o fixed to the object or to the hand. By Lemma 3.4, $sw \in \mathcal{X}_o(\nu_N)$ if and only if $sT_o^{-1}w \in \mathcal{X}(\nu_N)$, so a load about o can always be moved to the origin. We keep o explicit below, since the programs are written most simply with it; the support functions about o are $h_{\mathcal{X}_o(\nu_N)}(q) = \mu \int_P \|a + bJ(x - o)\|_2 \, d\nu_N$ and we write $\underline{h}_o$, $\overline{h}_o$, $\mathcal{X}_o^{\pm}$ for the corresponding bounds and sets.

Definition 9.1 (Load factors)**.** *For a nonzero load $w = (f, \tau) \in \mathbb{R}^2 \times \mathbb{R}$, expressed about o, the load factor of ν_N is $s_o^\star(\nu_N; w) \triangleq \max\{s \geq 0 : sw \in \mathcal{X}_o(\nu_N)\}$. The certified (lower) and the upper load factors are*

$$\underline{s}(w; y, \mathcal{M}) \triangleq \inf_{\nu_N \in \Pi_{\mathcal{M}}(y)} s_o^\star(\nu_N; w), \tag{55}$$

$$\overline{s}(w; y, \mathcal{M}) \triangleq \max\left\{s \geq 0 : sw \in \operatorname{cl} \mathcal{X}_o^+(y, \mathcal{M})\right\} \;\geq\; \sup_{\nu_N \in \Pi_{\mathcal{M}}(y)} s_o^\star(\nu_N; w). \tag{56}$$

The load w is carried by every compatible measure if $\underline{s}(w; y, \mathcal{M}) \geq 1$, and by none if $\overline{s}(w; y, \mathcal{M}) < 1$. Since $\mathcal{X}_o(\lambda\nu_N) = \lambda\mathcal{X}_o(\nu_N)$, a contact whose normal force is scaled by λ has load factors scaled by λ.

Proposition 9.2 (Certified and upper load factors)**.** *Let $w = (f, \tau) \neq 0$ and $o \in \mathbb{R}^2$.*

(i) For every measurement operator satisfying Assumption 3.9,

$$\boxed{\begin{aligned} &\underline{s}(w; y, \mathcal{M}) = \max\left\{s \geq 0 : sw \in \mathcal{X}_o^-(y, \mathcal{M})\right\} = \inf_{q:\, \langle q, w\rangle = 1} \underline{h}_o(q; y, \mathcal{M}), \\ &\overline{s}(w; y, \mathcal{M}) = \inf_{q:\, \langle q, w\rangle = 1} \overline{h}_o(q; y, \mathcal{M}). \end{aligned}} \tag{57}$$

(ii) On a taxel grid (Definition 5.1), for $\sigma \in \{-1, +1\}$ let

$$\boxed{\begin{aligned} V_\sigma \triangleq \mu \inf_{a \in \mathbb{R}^2,\, b \in \mathbb{R},\, u_i \in \mathbb{R}^2} \quad & \sum_{i=1}^{r} F_i \, \|a + J(u_i - b\, o)\|_2 \\ \text{subject to} \quad & a^\top f + b\tau = 1, \qquad \sigma b \geq 0, \\ & \sigma\,(u_i - b\,\mathrm{lo}_i) \geq 0, \quad \sigma\,(b\,\mathrm{hi}_i - u_i) \geq 0, \qquad i = 1, \ldots, r, \\ & \textstyle\sum_{i=1}^{r} F_i u_i = b\, m_1 \qquad (\mathcal{M}_T \text{ only}), \end{aligned}} \tag{58}$$

with $V_\sigma = +\infty$ if infeasible. Then $\underline{s}(w; y, \mathcal{M}) = \min\{V_{+1}, V_{-1}\}$ for $\mathcal{M} \in \{\mathcal{M}_F, \mathcal{M}_T\}$, and each V_σ is an SOCP.

(iii) Under the same assumptions, with the taxel corners v_{ik}, the upper load factor is the SOCP

$$\boxed{\begin{aligned} &\overline{s}(w; y, \mathcal{M}_F) = \min_{a,b,t} \; \mu \sum_{i=1}^{r} F_i t_i \quad s.t. \;\; a^\top f + b\tau = 1, \;\; t_i \geq \|a + bJ(v_{ik} - o)\|_2 \;\; \forall i, k, \\ &\overline{s}(w; y, \mathcal{M}_T) = \min_{a,b,y,z} \; \sum_{i=1}^{r} F_i y_i + z^\top m_1 \\ &\qquad\qquad s.t. \;\; a^\top f + b\tau = 1, \;\; y_i + z^\top v_{ik} \geq \mu \|a + bJ(v_{ik} - o)\|_2 \;\; \forall i, k, \end{aligned}} \tag{59}$$

and equivalently the primal SOCP

$$\overline{s}(w;y,\mathcal{M}) = \max_{s,\,\lambda_{ik}\geq 0,\,\rho_{ik}\in\mathbb{R}^2} s \quad s.t. \quad \textstyle\sum_k \lambda_{ik} = F_i, \quad \|\rho_{ik}\|_2 \leq \mu\lambda_{ik}, \tag{60}$$

$$\textstyle\sum_{i,k} \rho_{ik} = sf, \quad \sum_{i,k}(v_{ik}-o)\times\rho_{ik} = s\tau, \tag{61}$$

$$\textstyle\sum_{i,k} \lambda_{ik} v_{ik} = m_1 \quad (\mathcal{M}_T \text{ only}), \tag{62}$$

whose optimal λ is a most favorable closed-cell measure, with friction forces ρ_{ik} at its corners.

(iv) If $(a^\star, b^\star, u^\star)$ is optimal for the smaller of the two programs in (ii), let $c_i = u_i^\star/b^\star$ if $b^\star \neq 0$, and let (c_i) be any point loads with $c_i \in \overline{P_i}$ (and $\sum_i F_i c_i = m_1$ for $\mathcal{M}_T$) otherwise. Then $\hat{\nu}_N = \sum_i F_i \delta_{c_i}$ is a closed-cell measure with $s_o^\star(\hat{\nu}_N; w) = \underline{s}(w;y,\mathcal{M})$: the certificate is attained, by a compatible measure when each c_i lies in its own taxel.

(v) (Decision rule.) If $\underline{s} \geq 1$, every compatible normal force measure carries w. If $\overline{s} < 1$, none does: whatever the actual distribution, no admissible friction field balances w, and the contact slips (quasi-statically). If $\underline{s} < 1 \leq \overline{s}$, the measurement does not decide. At $\underline{s} = 1$ the load is carried but at the onset of sliding for the least favorable compatible measure; $\underline{s} > 1$ gives a strict margin. If $\mathcal{M} \succeq \mathcal{M}'$, then $\underline{s}(w;y',\mathcal{M}') \leq \underline{s}(w;y,\mathcal{M}) \leq s_o^\star(\nu_N^\star; w) \leq \overline{s}(w;y,\mathcal{M}) \leq \overline{s}(w;y',\mathcal{M}')$.

(vi) (Several contacts.) Let K contacts act on the same rigid object, with independent measurements y_k of their normal force measures ν_N^k, the same μ, and patches identified with subsets of a common plane (common coordinates x, torques about the common normal). Assume that the wrench components out of this plane are balanced without friction. The load w is then carried if $w \in \mathcal{X}_o(\nu_N^1) \oplus \cdots \oplus \mathcal{X}_o(\nu_N^K)$ (Minkowski sum). The certified and upper load factors of the grasp are

$$\boxed{\underline{s}_{\mathrm{grasp}}(w) = \inf_{q:\,\langle q,w\rangle=1} \sum_{k=1}^{K} \underline{h}_o(q;y_k,\mathcal{M}^k), \qquad \overline{s}_{\mathrm{grasp}}(w) = \inf_{q:\,\langle q,w\rangle=1} \sum_{k=1}^{K} \overline{h}_o(q;y_k,\mathcal{M}^k),} \tag{63}$$

and (ii)–(iii) hold with the objectives of the K contacts summed, a shared (a,b), and separate variables per contact. Since $\bigoplus_k \mathcal{X}_o^-(y_k) \subseteq \bigcap_{(\nu_N^k)} \bigoplus_k \mathcal{X}_o(\nu_N^k)$, $\underline{s}_{\mathrm{grasp}}$ is at least the load factor of the sum of the guaranteed sets of the individual contacts.

Proof sketch. (i) Fix ν_N. Since $\mathcal{X}_o(\nu_N)$ is convex, compact and contains 0, $\{s \geq 0 : sw \in \mathcal{X}_o(\nu_N)\} = [0, s_o^\star(\nu_N; w)]$ with $s_o^\star < \infty$. Hence $sw \in \mathcal{X}_o^-(y)$ if and only if $s \leq s_o^\star(\nu_N; w)$ for all compatible ν_N, which gives the first equality. A closed convex set is the set of points satisfying $\langle q, \cdot\rangle \leq h(q)$ for all q [4, Section 13]. For $\langle q, w\rangle \leq 0$ this holds for every $s \geq 0$, since $h_{\mathcal{X}_o(\nu_N)} \geq 0$. Therefore $s_o^\star(\nu_N; w) = \inf_{\langle q,w\rangle>0} h_{\mathcal{X}_o(\nu_N)}(q)/\langle q, w\rangle = \inf_{\langle q,w\rangle=1} h_{\mathcal{X}_o(\nu_N)}(q)$ by positive homogeneity (the constraint set is nonempty since $w \neq 0$). Exchanging the infima over ν_N and over q gives the second equality. For $\overline{s}$, the same argument applies to the closed convex set $\operatorname{cl}\mathcal{X}_o^+(y)$, which contains 0, is bounded by Assumption 3.9, and has support function $\overline{h}_o$ (Proposition 3.11(i)).

(ii) By (i) and Theorems 6.1 and 7.1, $\underline{s}(w;y,\mathcal{M})$ is the infimum of $\mu\sum_i F_i\|a + bJ(c_i - o)\|_2$ jointly over (a,b) with $a^\top f + b\tau = 1$ and over point loads $c_i \in \overline{P_i}$ (with $\sum_i F_i c_i = m_1$ for $\mathcal{M}_T$); for $\mathcal{M}_F$ the minimum over c_i is attained at the closed-form minimizers of Theorem 6.1. The objective is bilinear in (b, c_i). Split the feasible set by the sign of b and substitute $u_i = bc_i$. For $\sigma b > 0$, this is a bijection between the feasible c_i and the u_i with $u_i/b \in \overline{P_i}$, i.e., $\sigma(u_i - b\,\mathrm{lo}_i) \geq 0$ and $\sigma(b\,\mathrm{hi}_i - u_i) \geq 0$, and $\sum_i F_i u_i = b\,m_1$. Moreover $bJ(c_i - o) = J(u_i - b\,o)$, so the objective is preserved. For $b = 0$, the original objective is $\mu f_N\|a\|_2$ for every feasible (c_i), and feasible points exist. In the new variables, the box constraints force $u_i = 0$, and the objective is again $\mu f_N\|a\|_2$. The objective is a sum of norms of affine functions and the constraints are linear, so epigraph variables give an SOCP.

(iii) By (i) and Theorem 6.1, for $\mathcal{M}_F$, $\overline{s} = \inf \mu \sum_i F_i \max_k \|a + bJ(v_{ik} - o)\|_2$ over $\langle q, w \rangle = 1$, which is the stated program in epigraph form. For $\mathcal{M}_T$, the LP of Theorem 7.2 (with $g(v_{ik}; q)$ replaced by $\|a + bJ(v_{ik} - o)\|_2$) is feasible and bounded, so by LP duality $\overline{h}_o(q) = \min\{\sum_i F_i y_i + z^\top m_1 : y_i + z^\top v_{ik} \geq \mu \|a + bJ(v_{ik} - o)\|_2\}$; minimizing jointly over (q, y, z) gives the stated program, whose constraints are second-order cones. For the primal form: by Lemmas 5.5 and 5.6(ii), $\operatorname{cl} \mathcal{X}_o^+(y)$ is the union of the sets $\mathcal{X}_o$ of the corner-supported closed-cell measures, i.e., the projection of the (compact, convex) feasible set of (λ, ρ) onto the wrench $(\sum \rho, \sum (v - o) \times \rho)$.

(iv) By (i) and Lemma 5.5, $\underline{s}$ is also the infimum of $s_o^\star$ over the closed-cell measures, and $\hat{\nu}_N$ is one. With $q^\star = (a^\star, b^\star)$, by (i), $s_o^\star(\hat{\nu}_N; w) \leq h_{\mathcal{X}_o(\hat{\nu}_N)}(q^\star) = \mu \sum_i F_i \|a^\star + b^\star J(c_i - o)\|_2 = \underline{s}(w; y, \mathcal{M}) \leq s_o^\star(\hat{\nu}_N; w)$.

(v) Definitions, and Theorem 3.16 for the sets.

(vi) For fixed measures, $\bigoplus_k \mathcal{X}_o(\nu_N^k)$ is convex, compact, contains 0, and has support function $\sum_k h_{\mathcal{X}_o(\nu_N^k)}$. The argument of (i) applies. The compatible family is a product of the individual families, and the k-th term of the sum depends only on ν_N^k, so for fixed q the infimum (resp. supremum) of the sum is the sum of the infima (resp. suprema). For the inclusion, if $w_k \in \mathcal{X}_o^-(y_k)$ for each k, then $w_k \in \mathcal{X}_o(\nu_N^k)$ for every compatible ν_N^k, so $\sum_k w_k \in \bigoplus_k \mathcal{X}_o(\nu_N^k)$. □

Remark 9.3 (On Proposition 9.2).

- *Why not a grid of directions.* $\underline{h}_o(\cdot; y)$ *is an infimum of convex functions of* q*, and is not convex in general. Minimizing it over a finite set of directions gives a value* $\geq \underline{s}(w; y, \mathcal{M})$*, which may certify a load that is not guaranteed. Minimizing jointly over* q *and the point loads, after the change of variables* $u_i = bc_i$*, gives the exact value as a convex program. The upper bound* $\overline{h}_o(\cdot; y)$ *is convex (a support function), so* $\overline{s}$ *is a convex minimization from the start.*

- *Coarse case. With a single taxel and* $o = c$ *(*$\mathcal{M}_0$*), the constraint forces* $c_1 = c$*, and (ii) gives* $\underline{s} = \mu f_N / \|f\|_2$ *if* $\tau = 0$ *and* $\underline{s} = 0$ *if* $\tau \neq 0$ *(take* $a = 0$*,* $b = 1/\tau$*). The coarse measurement never certifies a torsional load about the CoP (Corollary 4.3).*

- *Membership tests for general operators. When* $\overline{h}$ *can be evaluated with a maximizing wrench (a support point of* $\mathcal{X}^+$*) but no closed form is available, e.g., through moment relaxations of the GMP, the distance from* w *to* $\operatorname{cl} \mathcal{X}^+$ *can be computed by the Frank–Wolfe method, whose linear minimization oracle is exactly this support point. For the taxel measurements, (iii) is exact and cheaper.*

- *Taxel boundaries. The values are exact for the taxel grid with its boundary convention (Lemma 5.5); other layouts are discussed in Remark 7.4.*

- *Scope. The certificate is quasi-static and applies to the measured state. In a feedback loop the load and the contact change between decisions, which requires a separate margin. In (vi), the assumption on the out-of-plane components matters: with a free object, offset normal forces at opposing contacts create a couple about an in-plane axis that must be balanced by friction, which the planar model does not account for.*

References

[1] L. C. Evans and R. F. Gariepy, *Measure Theory and Fine Properties of Functions*, Revised ed., CRC Press, Boca Raton, FL, 2015.

[2] J.-B. Lasserre, *Moments, Positive Polynomials and Their Applications*, Imperial College Press Optimization Series, vol. 1, Imperial College Press, London, 2010.

[3] G. Winkler, "Extreme points of moment sets," *Mathematics of Operations Research*, vol. 13, no. 4, pp. 581–587, 1988.

[4] R. T. Rockafellar, *Convex Analysis*, Princeton University Press, Princeton, NJ, 1970.